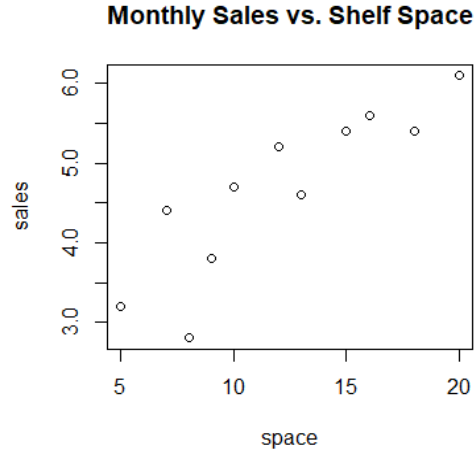


1.29

(a)



(b)

$n = 11$, $\sum_{i=1}^n x_i y_i = 662.4$, $\bar{x} = 12.091$, $\bar{y} = 4.655$. Hence,

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{n-1} = \frac{662.4 - 11(12.091)(4.655)}{10} = 4.33$$

The variance of the pressure is

$$s_x^2 = \frac{\sum_{i=1}^n x_i^2 - n(\bar{x})^2}{n-1} = \frac{1837 - 11(12.091)^2}{10} = 22.89$$

and the standard deviation is $s_x = \sqrt{22.89} = 4.78$.

The variance of the heights is

$$s_y^2 = \frac{\sum_{i=1}^n y_i^2 - n(\bar{y})^2}{n-1} = \frac{249.06 - 11(4.655)^2}{10} = 1.075$$

and the standard deviation is $s_y = \sqrt{1.075} = 1.04$.

$$r = \frac{\text{Cov}(x, y)}{s_x s_y} = \frac{4.33}{(4.78)(1.04)} = 0.87$$

The correlation coefficient is 0.87.

2.2

$$x^2 - 9 = (x - 3)(x + 3) = 0 \Rightarrow x = 3 \text{ or } x = -3 \Rightarrow A = \{-3, 3\}$$

$$x^2 - 4x + 3 = (x - 1)(x - 3) = 0 \Rightarrow x = 1 \text{ or } x = 3 \Rightarrow B = \{1, 3\}$$

$$A \cup B = \{-3, 1, 3\}$$

2.12

(a)

$$\binom{24}{3} = \frac{24!}{3!(24-3)!} = \frac{24!}{21!} = \frac{24 \cdot 23 \cdot 22}{3!} = 2,024$$

(b) All three members are women, or two members are women and one is a man.

$$\binom{4}{3} \binom{20}{0} + \binom{4}{2} \binom{20}{1} = \frac{4!}{3!(4-3)!} + \frac{4!}{2!(4-2)!} \cdot \frac{20!}{1!(20-1)!} = 4 + \frac{4 \cdot 3}{2} \cdot 20 = 124$$

(c)

$${}_{24}P_3 = \frac{24!}{(24-3)!} = \frac{24!}{21!} = 24 \cdot 23 \cdot 22 = 12,144$$

2.17

Define E as the event that a person can speak English, F as the event that a person can speak French, and G as the event that a person can speak German. Then

$$P(E) = 0.68, P(F) = 0.45, P(G) = 0.42, P(E \cap F) = 0.27, P(E \cap G) = 0.25, P(F \cap G) = 0.16, P(E \cap F \cap G) = 0.09$$

$$\begin{aligned} P(E \cup F \cup G) &= P(E) + P(F) + P(G) - P(E \cap F) - P(E \cap G) - P(F \cap G) \\ &\quad + P(E \cap F \cap G) = 0.68 + 0.45 + 0.42 - 0.27 - 0.25 - 0.16 + 0.09 \\ &= 0.96 \end{aligned}$$

2.22 The measurement is the birthday of each of the k people selected. Thus the number of possible outcomes is $(365)^k$. To complete the solution, we need to know in how many of these outcomes does some birthday appear at least twice. This is not an easy question to answer directly, but its complement is simple: In how many ways can k people all have different birthdays? It is the number of permutations of k birthdays selected from 365. If we let A be the event that no two people have the same birthday, then

$$P(A) = \frac{{}_{365}P_k}{(365)^k}$$

The event that at least two people have the same birthday is A^c . Therefore the answer is

$$P(A^c) = 1 - P(A) = 1 - \frac{{}_{365}P_k}{(365)^k}$$

(a)

$$1 - \frac{{}^{365}P_{23}}{(365)^{23}} = 1 - 0.493 = 0.507$$

(b)

$$1 - \frac{{}^{365}P_{30}}{(365)^{30}} = 1 - 0.294 = 0.706$$

(c)

$$1 - \frac{{}^{365}P_{50}}{(365)^{50}} = 1 - 0.030 = 0.970$$

2.31

(a) $P(B^c) = 1 - P(B) = 1 - 0.7 = 0.3$

(b) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.7 - 0.3 = 0.8$

(c)

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.7} = \frac{3}{7} = 0.43$$

2.36

Let W denote the event that Brian's team wins and Y denote the event that the pitcher pitches in a game. Then $P(W) = 0.55$, $P(Y) = 0.25$ and $P(Y \cap W) = 0.18$.

(a)

$$P(Y|W) = \frac{P(Y \cap W)}{P(W)} = \frac{0.18}{0.55} = \frac{18}{55} = 0.327$$

(b)

$$P(W|Y) = \frac{P(Y \cap W)}{P(Y)} = \frac{0.18}{0.25} = \frac{18}{25} = 0.72$$

(c)

$$P(W^c \cup Y^c) = P((W \cap Y)^c) = 1 - P(W \cap Y) = 1 - 0.18 = 0.82$$

(d)

$$\begin{aligned} P(Y^c \cap W^c) &= P((Y \cup W)^c) = 1 - P(Y \cup W) = 1 - [P(Y) + P(W) - P(Y \cap W)] \\ &= 1 - (0.25 + 0.55 - 0.18) = 0.38 \end{aligned}$$